Ishani Cheshire

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Data scientist with expertise in machine learning, statistical modeling, and multilingual NLP. Experienced in building predictive models, designing data pipelines, and translating complex analyses into actionable insights. Strong background in Python, SQL, deep learning frameworks, and cloud platforms (AWS). Outstanding Graduate Student Instructor Award winner & Teaching Effectiveness Award winner, UC Berkeley 2024–2025.

EDUCATION

University of California, Berkeley — **Aug 2024 – May 2026**
Master's in Data Science | GPA: 4.0 — *Berkeley, CA*

University of California, Berkeley — **Aug 2019 – Dec 2023**
B.A. in Astrophysics; B.A. in Physics — *Berkeley, CA*

TECHNICAL SKILLS

Languages: Python, SQL, R, Lisp/Scheme, LaTeX, Java
Databases: PostgreSQL, Neo4j, MongoDB, Redis
Platforms: AWS SageMaker, EC2, DigitalOcean
Viz: Tableau, Observable, Altair, matplotlib
ML/AI: PyTorch, TensorFlow, Keras, scikit-learn, pandas, scipy, Hugging Face transformers

WORK EXPERIENCE

LLM-Augmented Web Scraping @ Briza — **May 2025 – Aug 2025**

- Designed a GPT-4 augmented ChromeDriver pipeline to retrieve grant opportunity data from curated websites, dynamically updating a RAG database as new sources were added.
- Fine-tuned an Ollama LLM to advise on public policy proposals, grounding outputs in research, grant data, and demographic context gathered via LLM-based web scraping.

UC Berkeley Astronomy — Head Graduate Student Instructor — **Aug 2024 – Jan 2026**

- Served as Head GSI for upper division courses, organizing weekly instructor meetings to align a team of tutors and acting as primary contact with faculty; proactively identified struggling students via performance data and held weekly check-ins.
- Designed an innovative team formation system for group projects — recognized with the Teaching Effectiveness Award and Outstanding GSI Award (2024–2025).

Chalmers Astrophysics Research Fellowship — **May 2022 – May 2023**

- Analyzed spectral data of >1,000 stars across three nearby open clusters using GAIA DR3; identified >600 pre-main sequence stars and submitted first-author findings to A&A (presented at AAS 241 and From Stars to Galaxies II).

NSF REU @ Cornell University — **May 2023 – Aug 2023**

- Built an image processing pipeline for STEM microscopy data; quantified molecular polarization patterns across unit cells and presented findings at the 2023 Sigma Xi Research Symposium.

PROJECTS

ArcRadius — Lead ML Engineer — **Jan 2026 – May 2026**
LLM-Powered LGBTQ+ Policy Navigator | Master's Capstone — *UC Berkeley*

- Fine-tuned a LegalBERT classification pipeline to label state legislation as supportive or harmful to LGBTQ+ youth; diagnosed and resolved model underperformance, driving improvement from 75% to 88% F1 through targeted retraining.
- Constructed graphRAG-augmented LLM summaries of bills, adding context of the policy landscape to support reliability.
- Designed and executed the full data pipeline — ingesting legislative data from LegiScan's API into a dynamic graph RAG database, restructuring nested JSON into tabular format, and handling class imbalance across relevant vs. irrelevant bills.
- Built a letter-generation feature using a fine-tuned LLM to produce personalized advocacy letters opposing anti-LGBTQ+ legislation, scoping the feature from user need through prompt design, evaluation, and integration.

Retrieval-Augmented Generation (RAG) System for GenAI Knowledge Retrieval — **Jan 2026 – May 2026**

- Built an end-to-end RAG pipeline for unstructured GenAI documents using LangChain, Qdrant, vector databases / embedding-based retrieval, and MMR retrieval to support engineering and marketing question-answering workflows.
- Improved answer quality and reduced hallucination risk through heuristic query rewriting, chunking/retrieval tuning, and persona-specific prompt pipelines, with explicit safeguards for grounded responses.
- Designed a multi-metric evaluation framework for retrieval relevance, groundedness, and response quality, using experimental results to identify retrieval bottlenecks and recommend a limited internal rollout.

Multilingual Hate Speech Detection (Deep Learning) — **May 2025 – Aug 2025**

- Fine-tuned mBERT, RemBERT, and XLM-RoBERTa to detect hate speech across English, Arabic, and French; achieved 95.5% test F1 for binary detection and 85.5% for a five-output multi-task model (+25–35% over baseline).
- Designed a hierarchical multi-task architecture with five prediction heads; performed data cleaning, class balancing, and stratified splitting on 30K+ records; generated LIME interpretability analyses for stakeholder transparency.